FUNDAMENTALS OF GENERATIVE AI

❖ Generative AI: -

Generative AI is a type of artificial intelligence that can **create new content**—instead of just analyzing or classifying existing data.

- Examples: -

  - ➢ ChatGPT → generates answers, essays, code
  - ➢ DALL·E → creates images from text
  - ➢ GitHub Copilot → writes code

- How it works: -

  1. AI is trained on a large dataset (text, images, etc.)
  2. It learns patterns and relationships
  3. When you give a prompt, it predicts and generates new content

  - Most modern generative AI uses models like:
    - ➢ Neural Networks
    - ➢ Transformers (like GPT models)

- Types of generative AI: -

  - ➢ Text generation → Chatbots, content writing
  - ➢ Image generation → AI art, design
  - ➢ Audio generation → Voice cloning, music
  - ➢ Code generation → Auto coding tools

- Real Life Use Cases
  - ➢ Writing emails, blogs, reports
  - ➢ Creating images/designs
  - ➢ Building chatbots

❖ Foundation Model: -

A Foundation Model is a large AI model trained on huge amounts of data (text, images, etc.) that can be used for many different tasks without building a model from scratch.

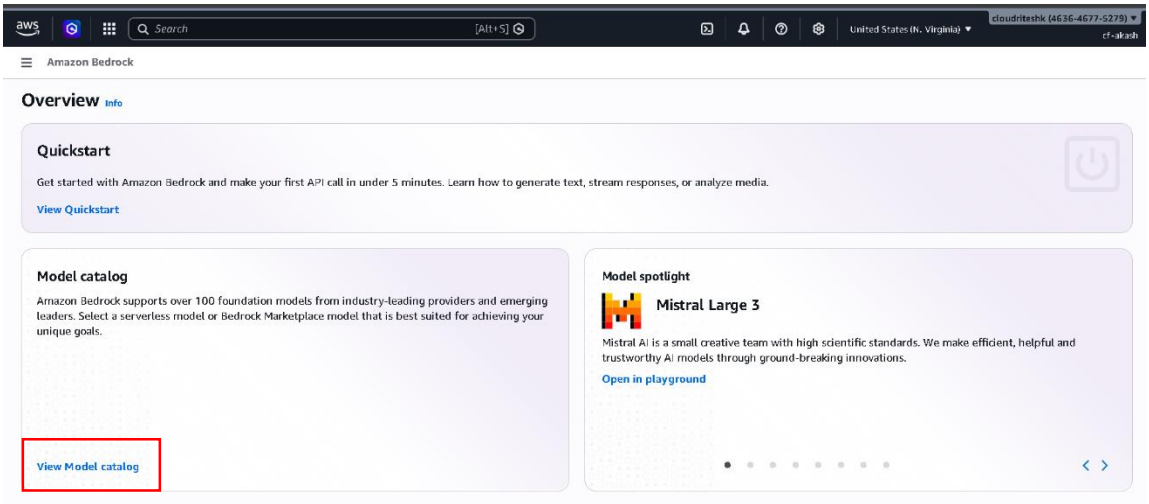
- There is broad categorization of models :-

  - **Text Generation.**

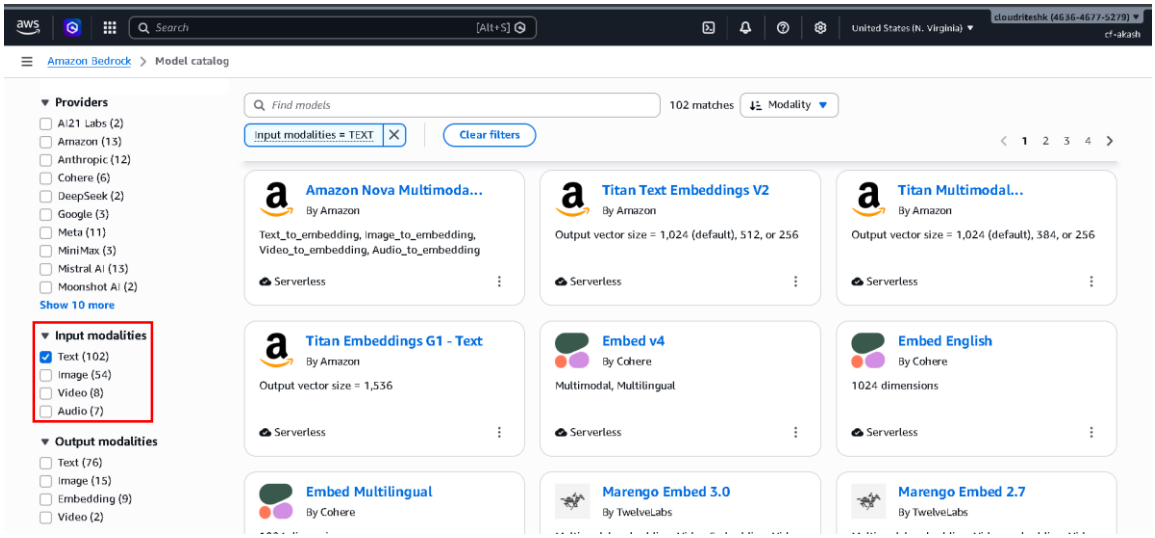
    - ➢ If we deals with text generation then it is usually called as a **Large Language Model (LLM's)**
    - ➢ Example: - GPT 4, Llama 3
    - ➢ Use Cases: - Text Summarization, Text Generation.

❖ **Now Let's see from Amazon Bedrock Console what are some of the common text generation Models: -**

- Go to your Amazon console and Search Amazon Bedrock.



- Click on the view Model Catalog



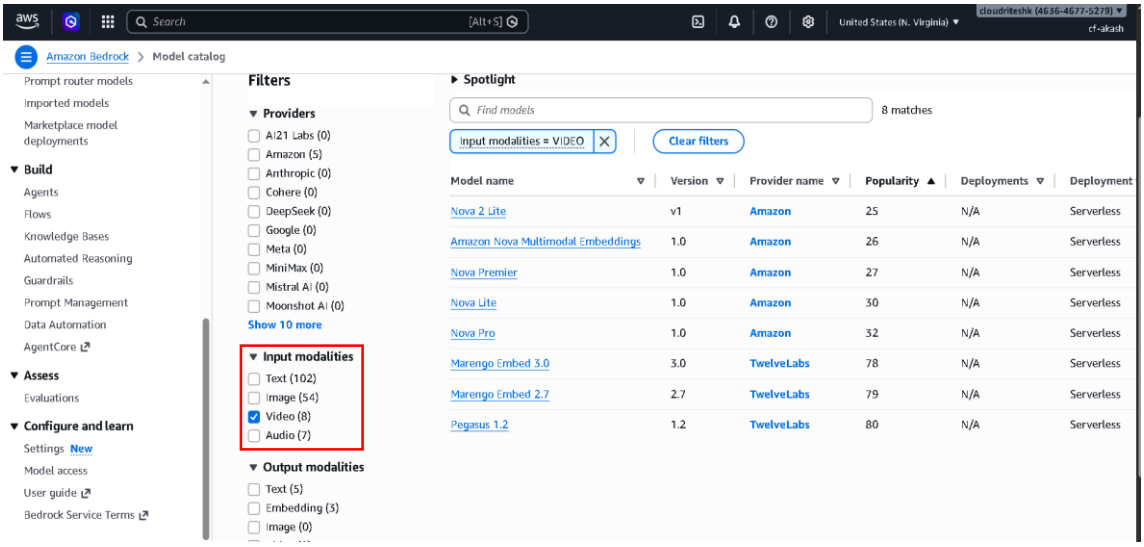
- Select text in modalities

- Now you see there are all the models that are focused on text and usually called as large language.

○ **Image Generation: -**

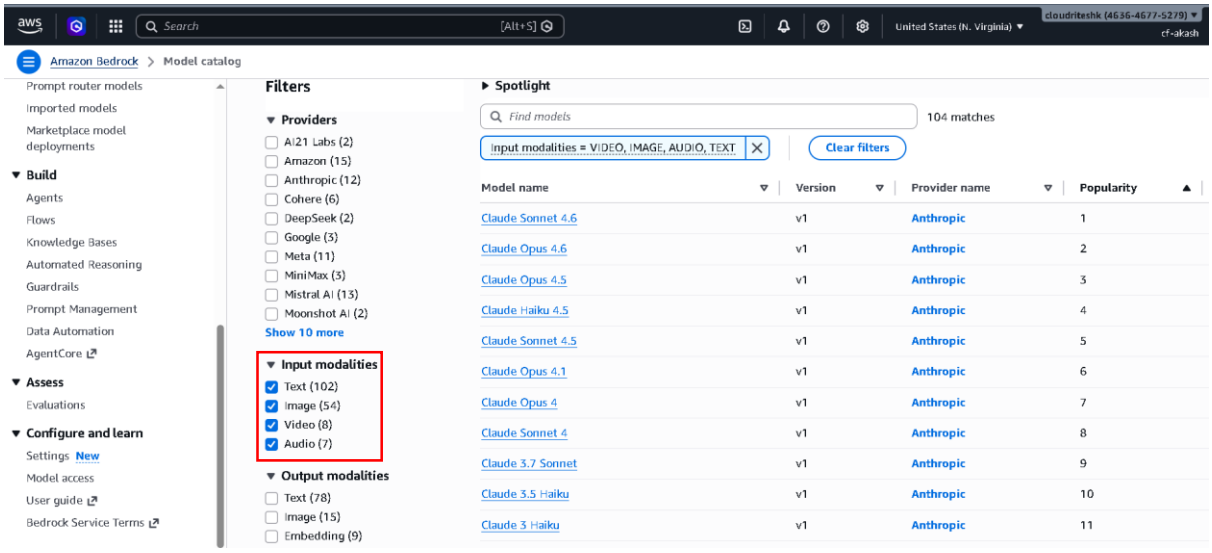
➢ Image generation is a type of generative AI where a model creates new images from text prompts or data**.**
➢ It is usually knowing as Image Generation Model or Diffusion Model.
➢ Example: - stable Diffusion (Amazon Bedrock), Dalle three (Open AI)

❖ **Now go to Amazon Bedrock and see: -**



- These are the Models.

- Multi Model: -
  - A multimodal model is an AI model that can understand and process multiple types of data together, such as:
    - Text
    - Audio
    - Video
    - Image
  - Example Claude3 – Multi Model



- Some Models that work on multiple Types.

❖ Vectors: -

- Why Do We Need Vectors: -

  - ➢ Computers don't understand text, images, or audio directly. They understand only numbers.
  - ➢ So we convert everything into vectors (arrays of numbers).

- What are Vectors: -

  - ➢ Vectors are mathematical representation of words, sentences and documents.
  - ➢ Vectors are represented by list of numbers or sequence of numbers.

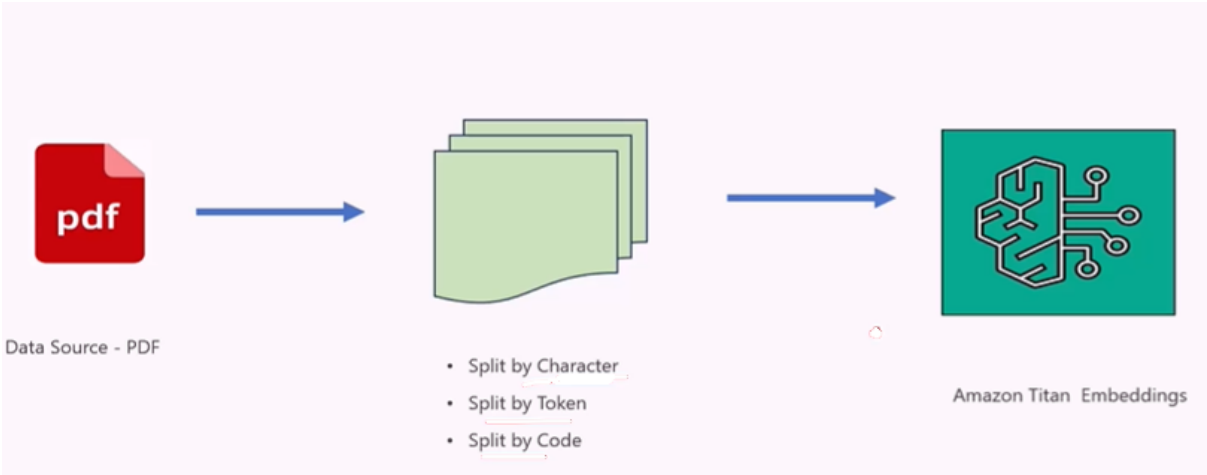
❖ Embedding Models: -

An embedding model is an AI model that transforms data into numerical vectors so that machines can understand and compare meaning.

Text / Image → Embedding Model → Vector (numbers)

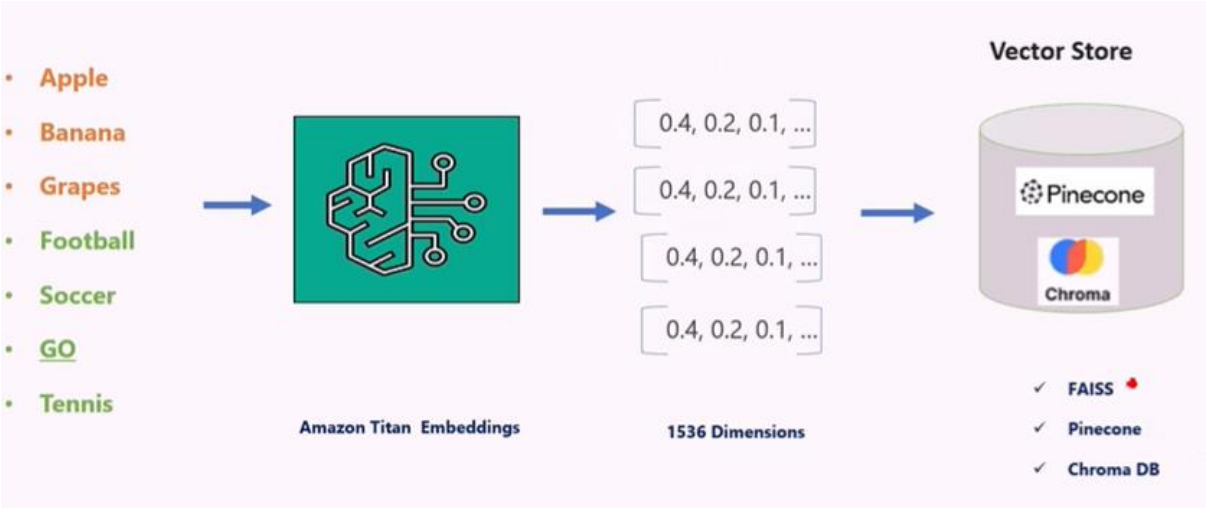
❖ Data Chunking: -

Data chunking is the process of dividing data into manageable pieces (chunks) so AI can process it efficiently.



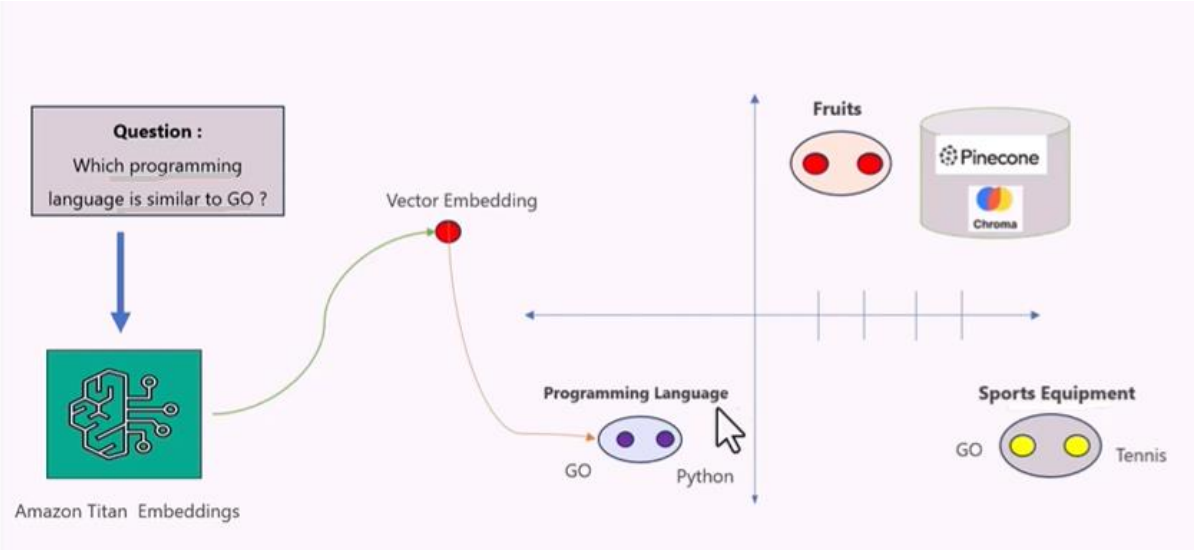
❖ Vectors Store: -

A vector store is a database that stores embedding's and enables fast similarity search for AI applications.



❖ Vector Search: -

Vector search is the process of finding similar items by comparing their vector representations using similarity measures.

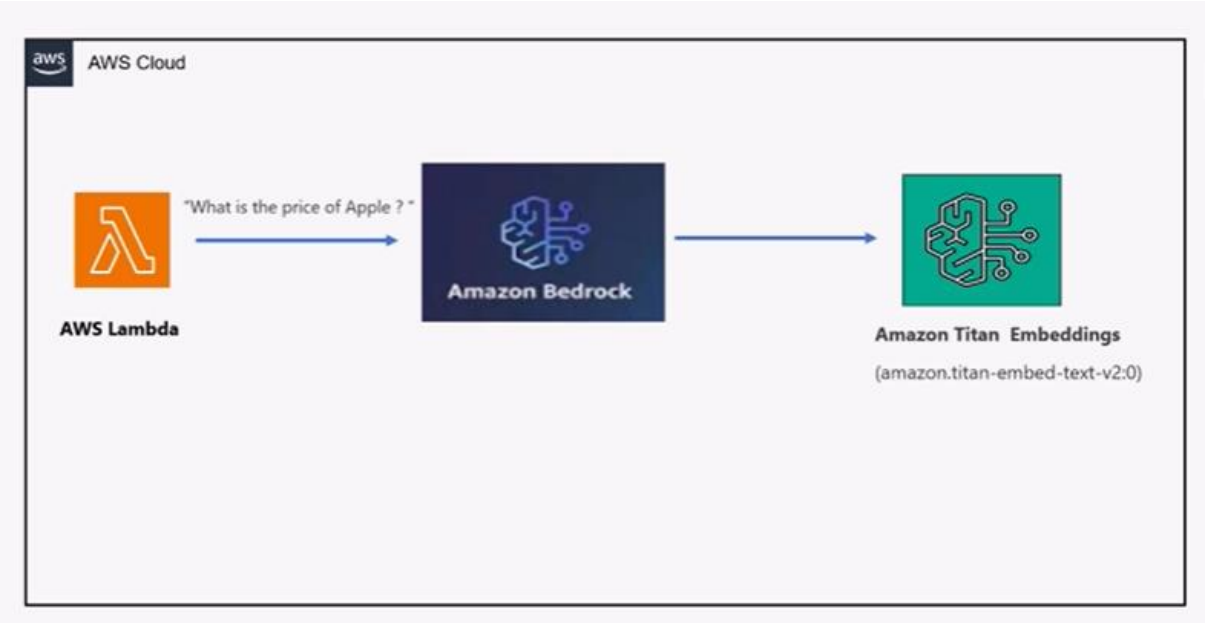


There are some Algorithm uses in Vector Search: -

- Cosine Similarity
- K- Nearest Neighbor

❖ Vector and Embedding's In AWS Cloud: -

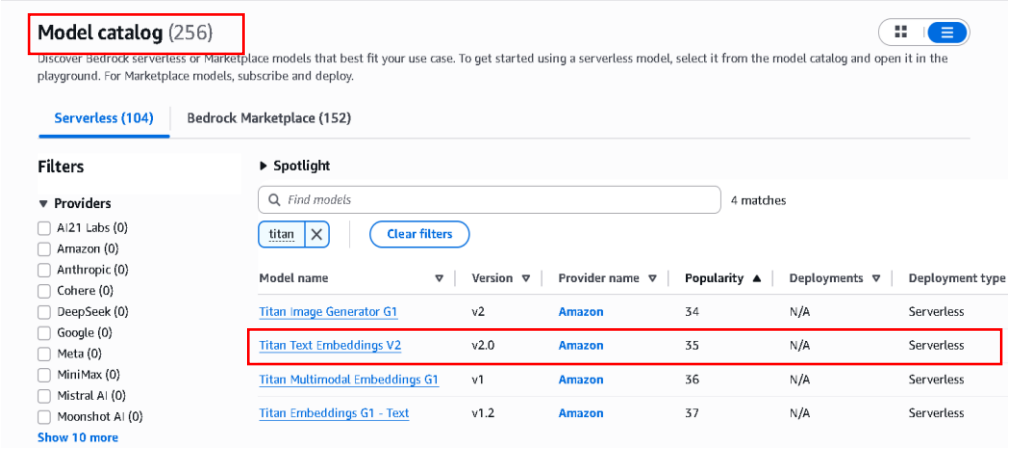
- So what we will do we have the AWS Cloud.



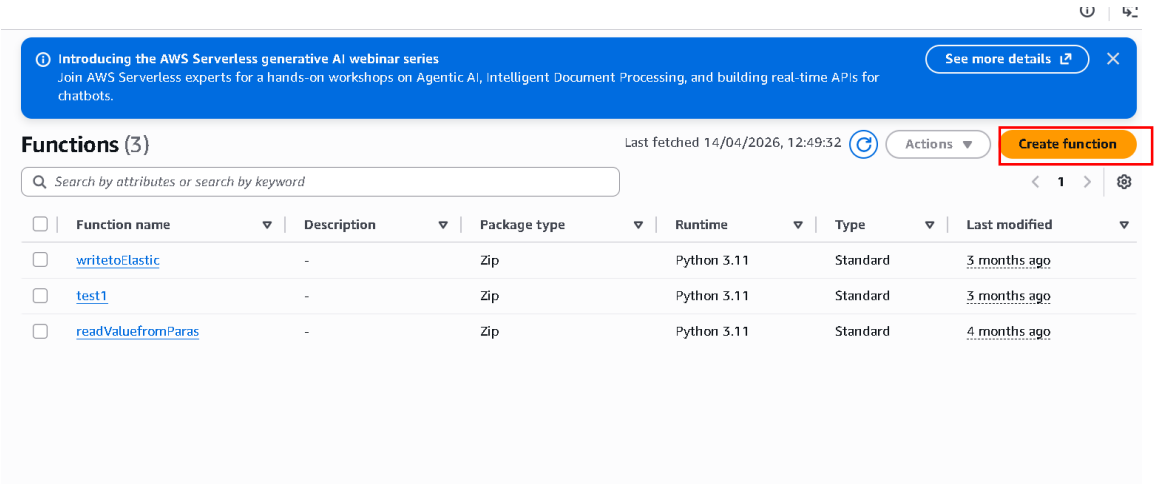
- With AWS cloud we create Lambda function.
- And through the lambda function we are going to send some prompts (What is the price of Apple?).
- Then we are going to send this word of sentences to Amazon Bedrock
- And Amazon bedrock behind the scenes is going to send this request to Amazon Titan Embedding's.
- And in particular we are going to use Amazon Dot titan Embed text version 2.0 Model.
- So once the request is going to Amazon titan embedding model.
- Its going to create a vector embedding of that particular model and send back the response to the lambda function.

❖ Now we are going to build this: -

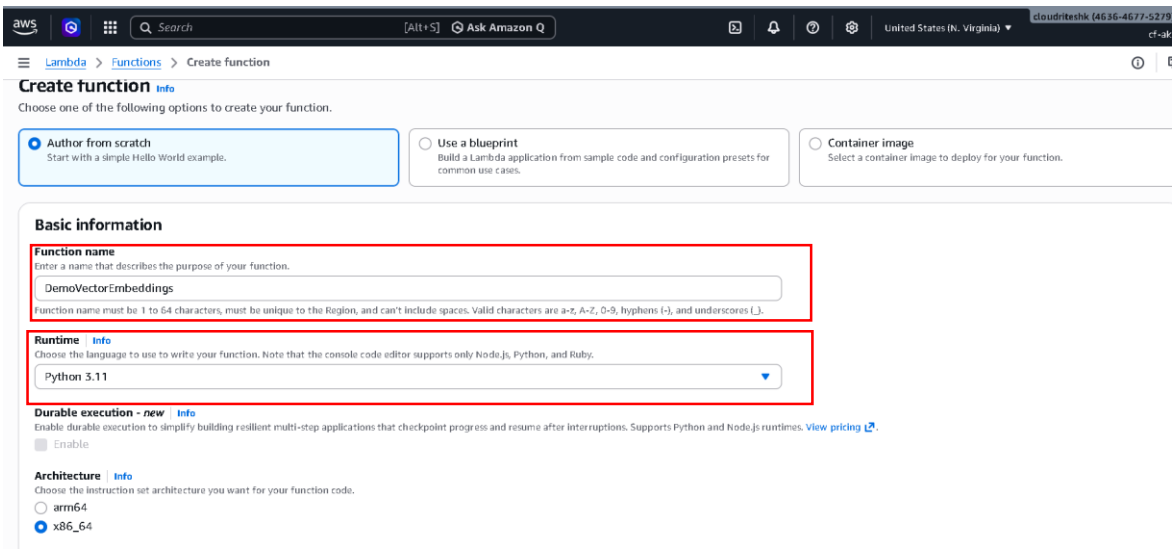
- Firstly go to your console and search Amazon Bedrock.
- Then go in view model catalog
- And search Titan Text Embeddings V2 in search bar.



- Open this Titan Text Embeddings V2
- Now open Another AWS console in new tab and Search Lambda in search bar.



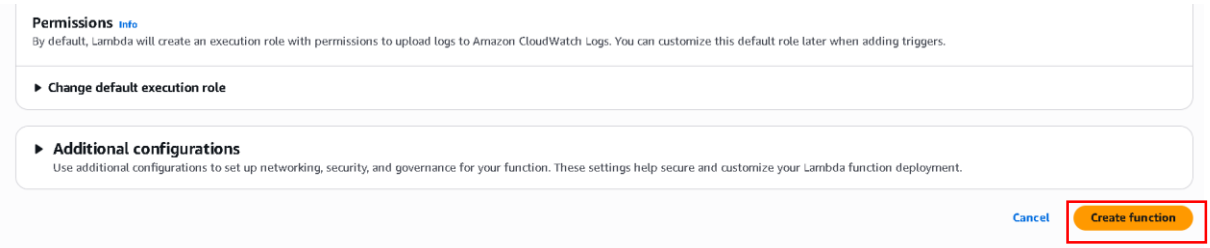
- Click on the create function.



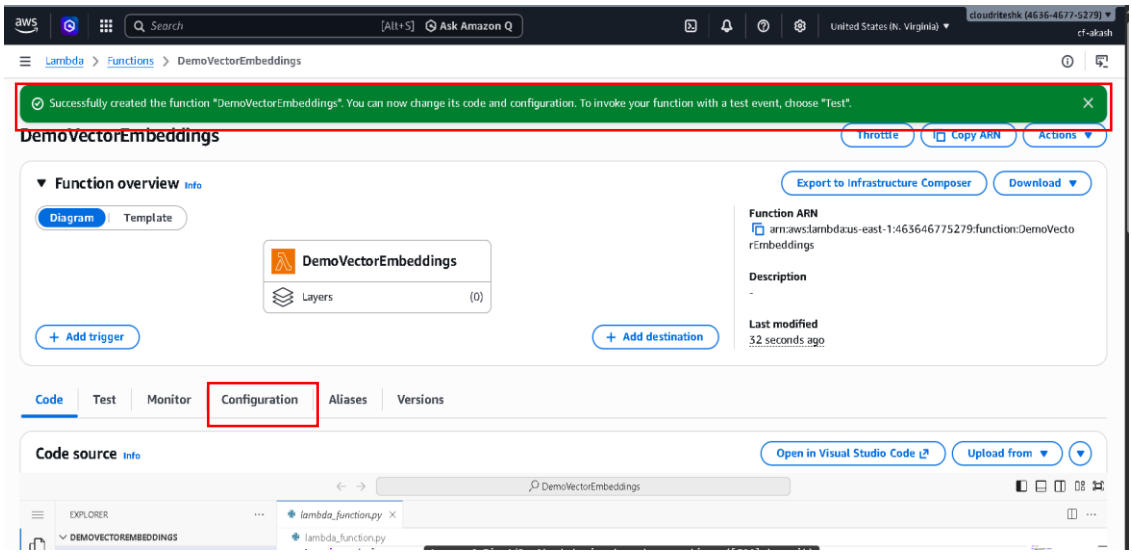
- choose author from scratch.

- Write function Name

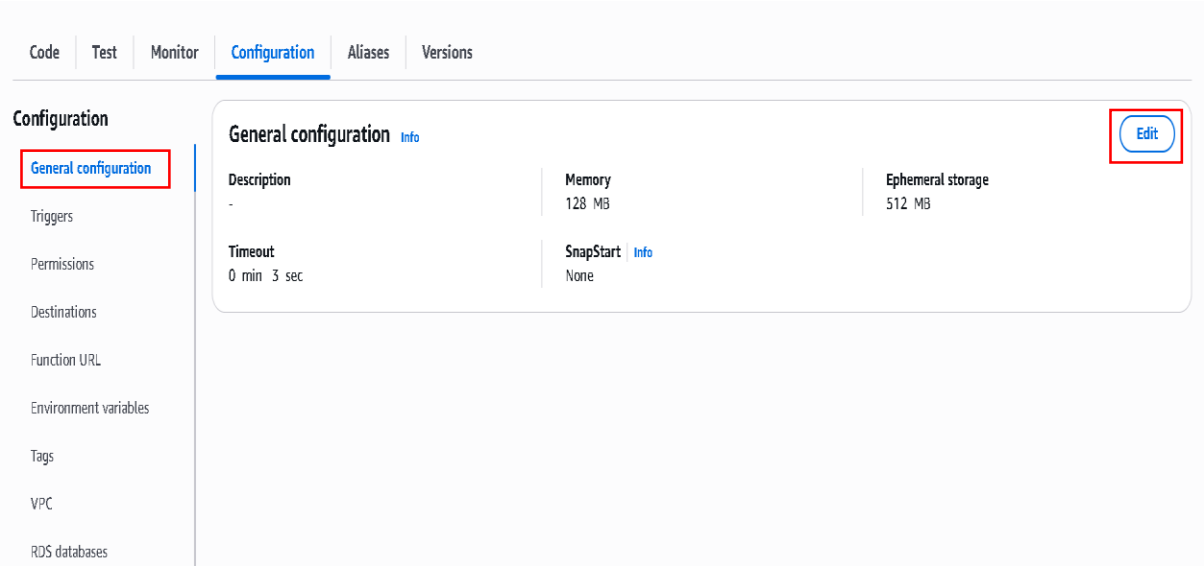
- And choose Runtime (Programing language) I choose Python 3.11



- Now click on create Account.

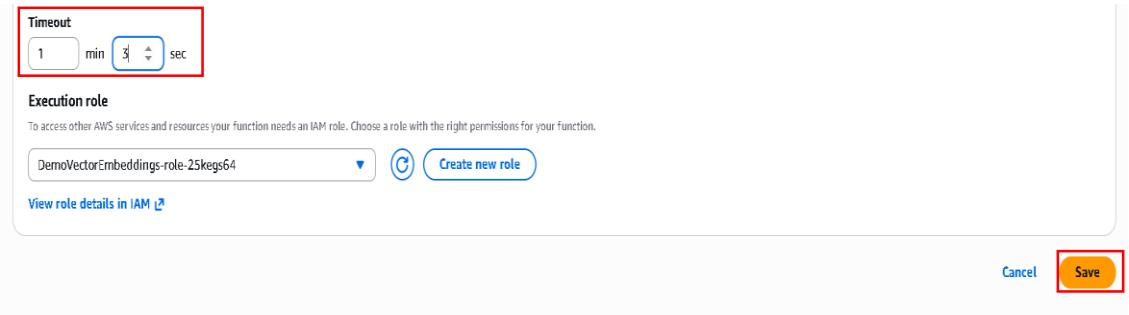


- So our lambda function is created.
- And now you go on configuration
- And in configuration go general configuration section.
- And click on Edit.

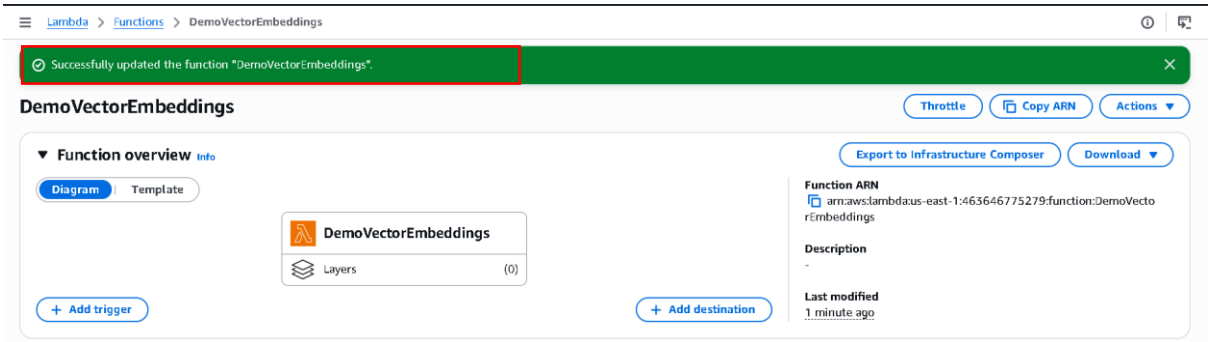


- The default timeout for lambda function is 3 second

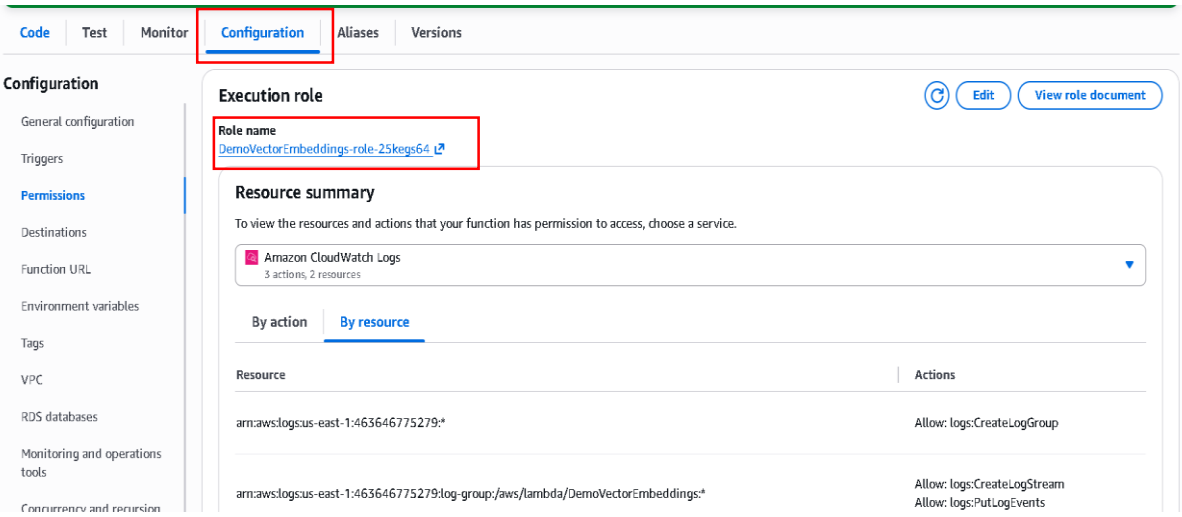
- So I will make it 1 min 3 sec because it might have to do some processing.



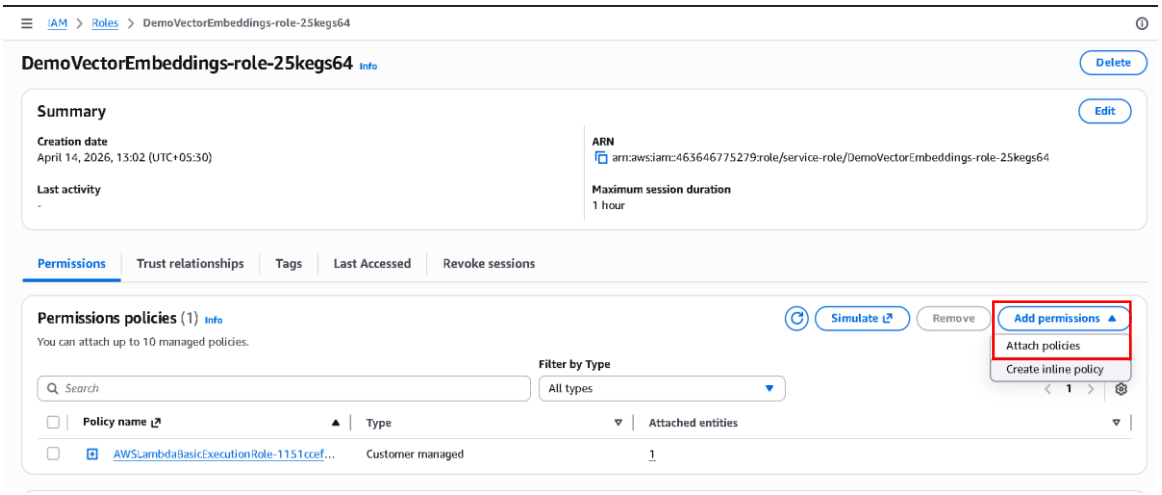
- And click on save.



- Function is successfully updated.
- Now Lambda Function also requires access to Amazon Bedrock.
- So we'll give it the required permissions.
- So go to configuration and click on Permission

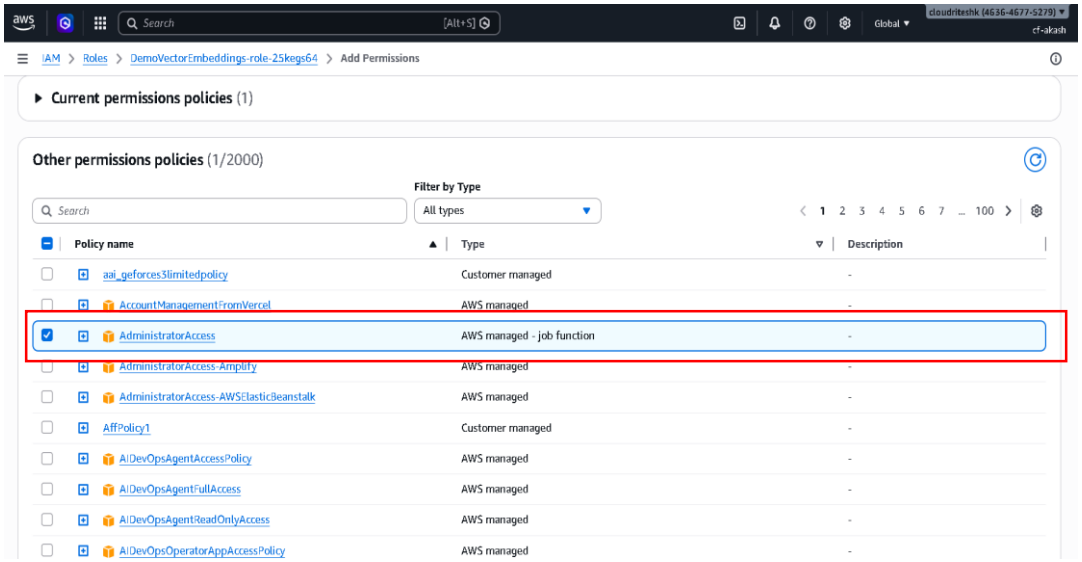


- Now click on role name
- And then in role name click on ADD Permissions
- In Add Permission click Attach Policies.

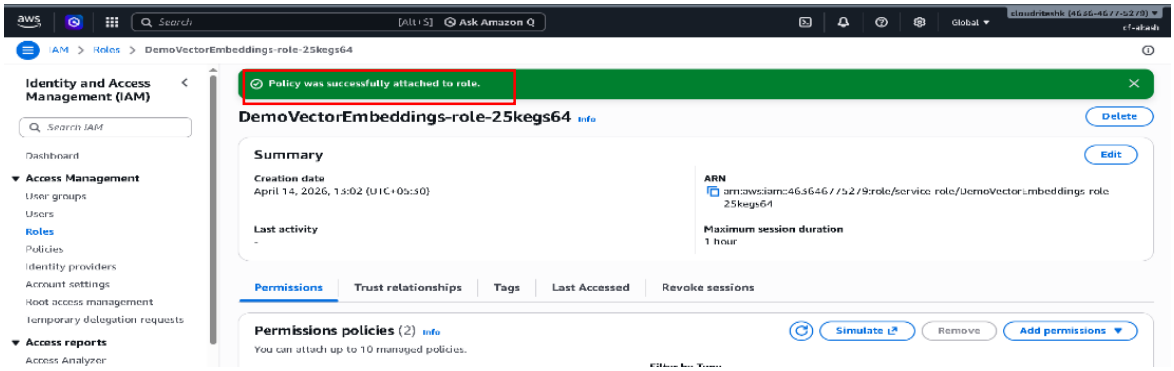


- So we'll give admin access

- This is not done in production Environment but for this demo we'll just give it admin access.



- Now just click on add Permissions.

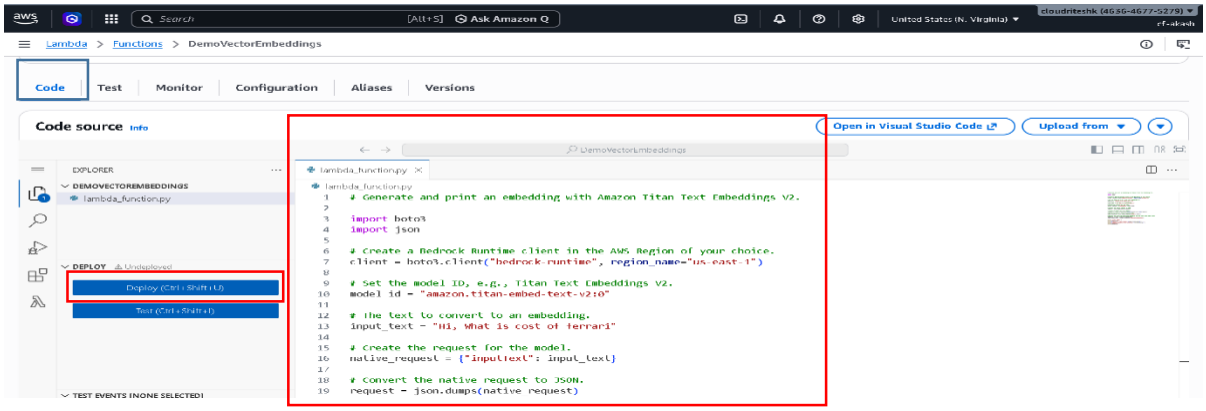


- Now Policy was successfully attached to role.
- Go back on lambda function tab and click on code.
- And paste the below code in the code section.

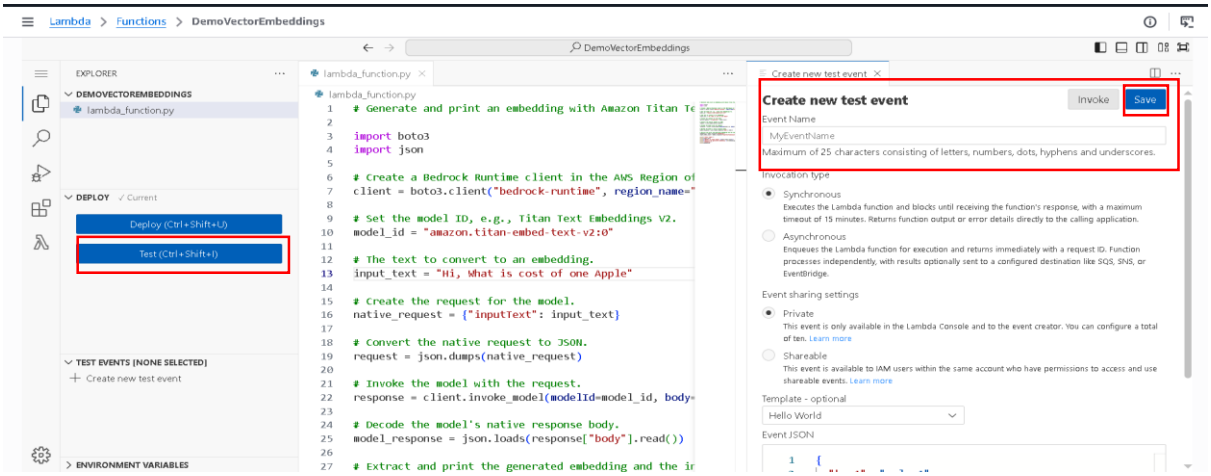
```
import boto3
import json

def lambda_handler(event, context):

    # 🔷 Step 1: Create Bedrock client
    client = boto3.client("bedrock-runtime", region_name="us-east-1")

    # 🔷 Step 2: Model ID (Titan Embeddings V2)
    model_id = "amazon.titan-embed-text-v2:0"

    # 🔷 Step 3: Get input text (dynamic or default)
    input_text = event.get("text", "What is the price of apple?")

    # 🔷 Step 4: Create request
    request_body = {
        "inputText": input_text
    }

    # 🔷 Step 5: Convert to JSON
    request = json.dumps(request_body)

    # 🔷 Step 6: Call Bedrock model
    response = client.invoke_model(
        modelId=model_id,
        body=request
    )

    # 🔷 Step 7: Read response
    response_body = json.loads(response["body"].read())

    embedding = response_body["embedding"]
    token_count = response_body["inputTextTokenCount"]

    # 🔥 Step 8: Print logs (CloudWatch)
    print("===== DEBUG LOGS =====")
    print("Input Text:", input_text)
    print("Token Count:", token_count)
    print("Embedding Size:", len(embedding))
    print("First 10 Embedding Values:", embedding[:10])

    # 🔷 Step 9: Return response (avoid full embedding)
    return {
        "statusCode": 200,
        "body": {
```

```
            "input": input_text,
            "token_count": token_count,
            "embedding_size": len(embedding),
            "sample_embedding": embedding[:10]   # only sample
        }
    }
```

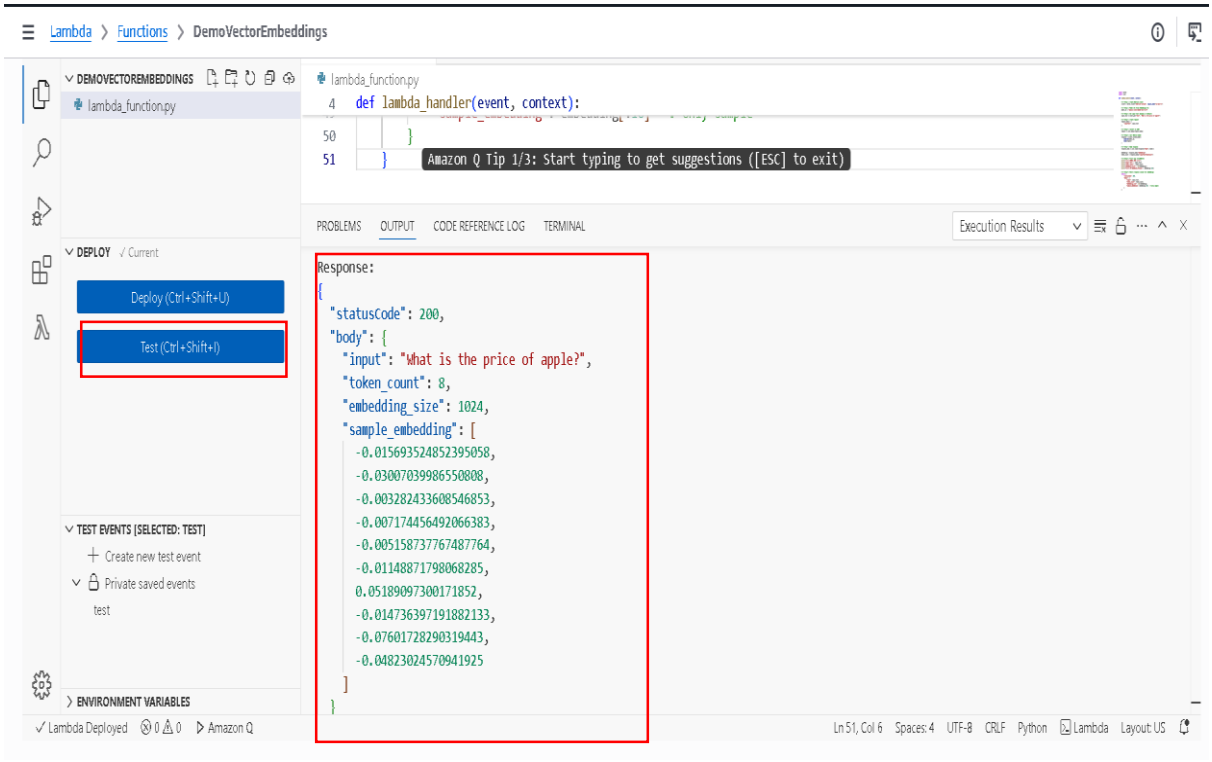
- Now click on Deploy



- Our code is deployed
- Now click on Test
- A popup window comes select create new test event



- Write event name (like test) and click on save.
- And then again click on test button.
- Your function runs and give output.


Lambda > Functions > DemoVectorEmbeddings
DEMOVECTOREMBEDDINGS
lambda_function.py
def lambda_handler(event, context):
Amazon Q Tip 1/3: Start typing to get suggestions ([ESC] to exit)
DEPLOY
Deploy (Ctrl+Shift+U)
Test (Ctrl+Shift+I)
TEST EVENTS [SELECTED: TEST]
Create new test event
Private saved events
test
ENVIRONMENT VARIABLES
PROBLEMS
OUTPUT
CODE REFERENCE LOG
TERMINAL
Execution Results
Response:
{
"statusCode": 200,
"body": {
"input": "What is the price of apple?",
"token_count": 8,
"embedding_size": 1024,
"sample_embedding": [
-0.015693524852395058,
-0.03007039986550808,
-0.003282433608546853,
-0.007174456492066383,
-0.005158737767487764,
-0.01148871798068285,
0.05189097300171852,
-0.014736397191882133,
-0.07601728290319443,
-0.04823024570941925
]
}
Lambda Deployed
Amazon Q
Ln 51, Col 6
Spaces: 4
UTF-8
CRLF
Python
Lambda
Layout: US